N53 K220 S3 root batching: valid run record

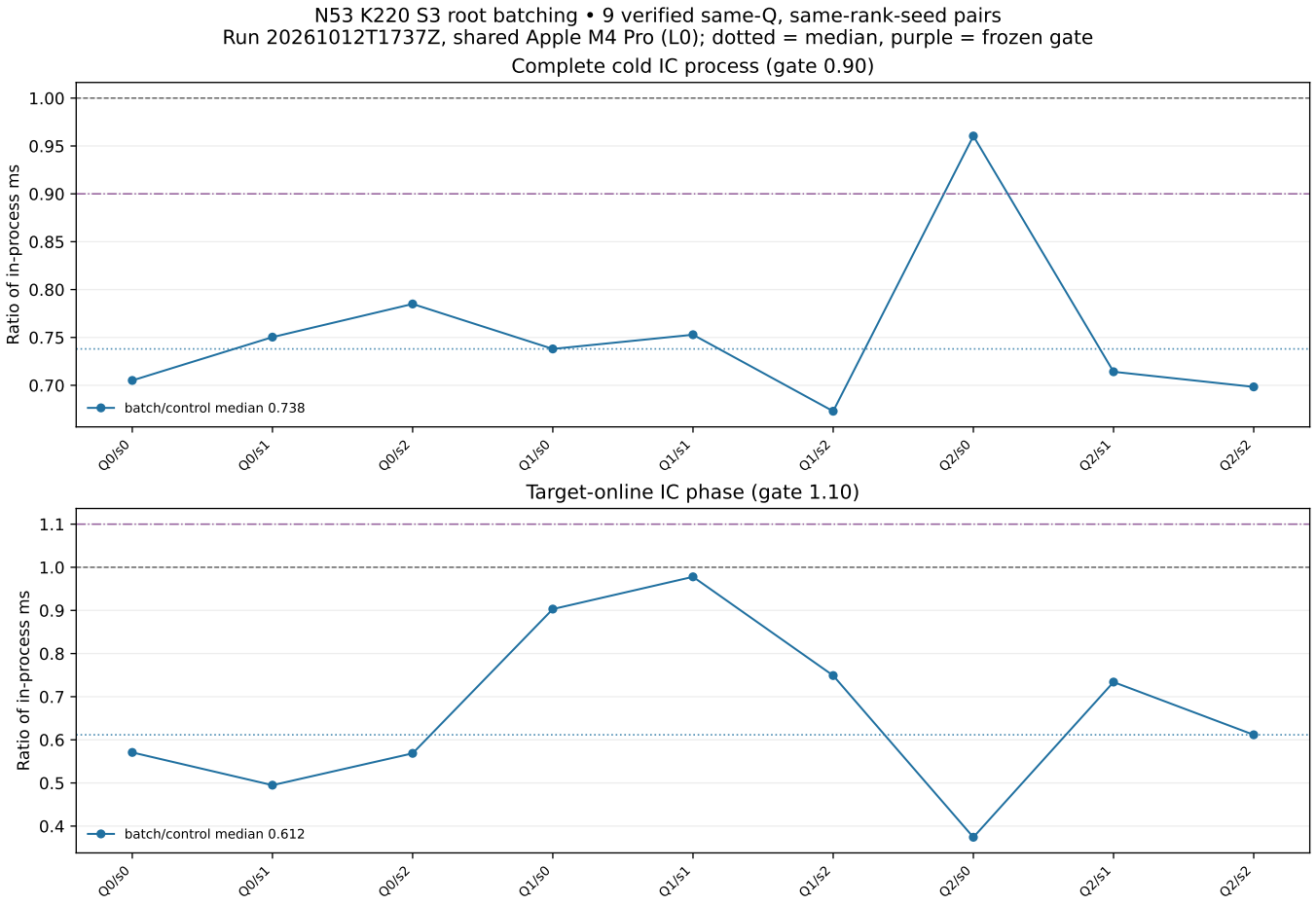
2026-10-12 17:37 UTC; 27 frozen cells; run 20261012T1737Z

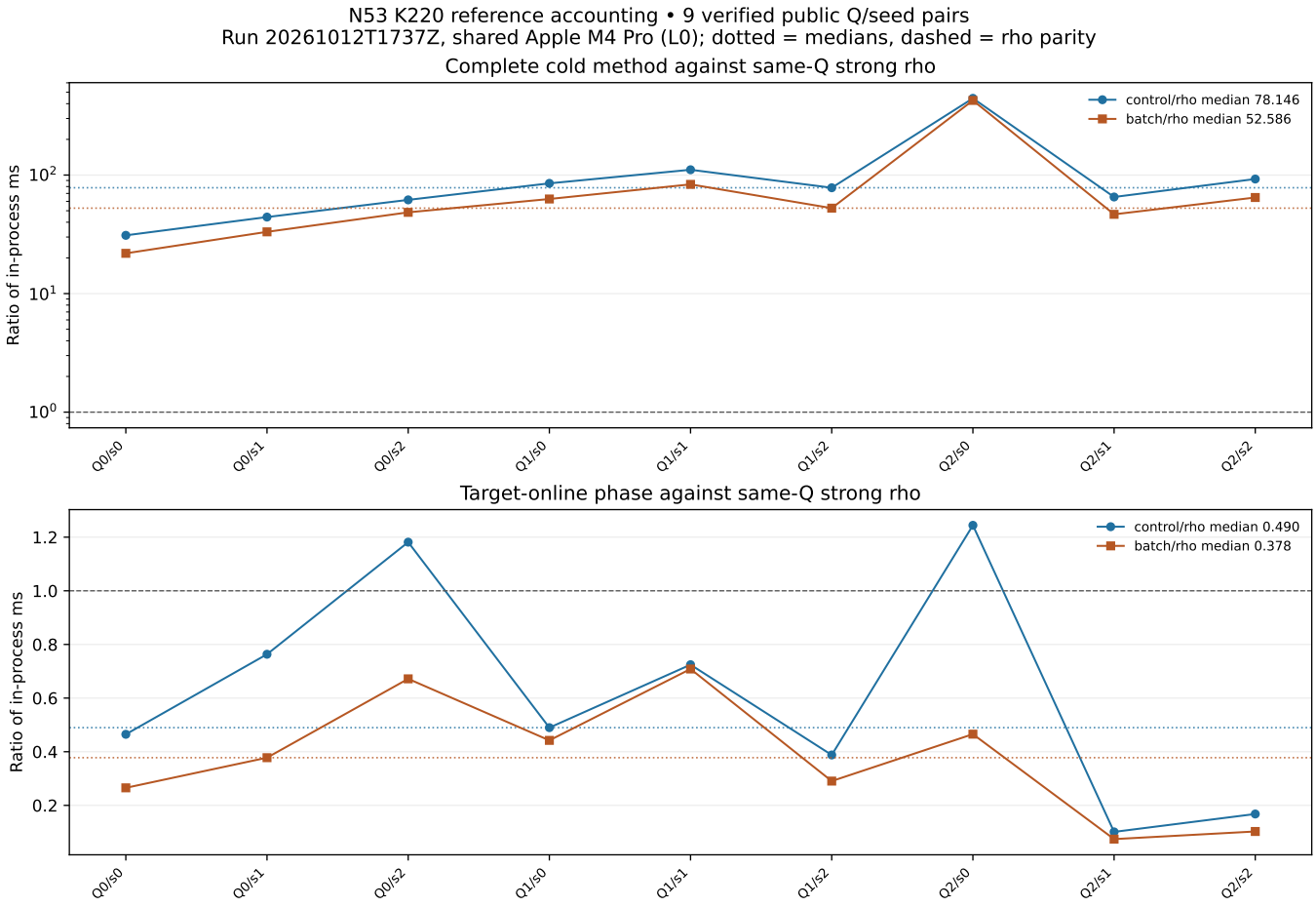
Identity and host. Curve EC1N53Ce0hb097de99be9a, prime subgroup 21,044,858,204,113; K220 fixed 23,320-point base. Three public Q, three rank seeds per Q, three strong-rho seeds. Source and binary hashes: `../../FROZEN.json`. Apple M4 Pro, macOS 26.6, Rust 1.93.1 release; shared-host L0.

Execution and correctness. All 27 process statuses are `COMPLETED` within the 90-second and 16-GiB caps. Native analysis and independent replay verified all nine control/candidate pairs: rank 220/220, 220 accepted independent relations per IC arm, same selected rank trace, base dump, root table, consumed root count, target witness and scalar. The 9 rho arms verified the same public Q. Raw records: `../../runs/`; native analysis: `../../ANALYSIS.json`; native aggregate: `../../RESULT_DATA.json`.

Same-Q ratio, in-process ms	Median	Target-block 95% interval
Batch/control, cold IC	0.737998	[0.714128, 0.750321]
Batch/control, target online	0.611643	[0.568613, 0.903106]
Control/strong rho, cold	78.145833	[44.219726, 92.601592]
Batch/strong rho, cold	52.585601	[33.179008, 64.667877]
Control/strong rho, online	0.489747	[0.167802, 0.763394]
Batch/strong rho, online	0.377771	[0.102635, 0.442294]

The intervals are 10,000 deterministic resamples of three target blocks, retaining each block's three rank seeds. These are descriptive timing summaries, without host-isolation admission or common-operation S calibration. Source-level rank-root totals over nine cells: 80,790,012 versus 763,167 inversions; 1,211,850,180 versus 902,063,394 multiplication calls; 242,370,036 versus 39,684,684 squaring calls (control versus batch).


N53 K220 S3 root batching • 9 verified same-Q, same-rank-seed pairs
Run 20261012T1737Z, shared Apple M4 Pro (L0); dotted = median, purple = frozen gate
Complete cold IC process (gate 0.90)
Ratio of in-process ms
1.00
0.95
0.90
0.85
0.80
0.75
0.70
batch/control median 0.738
Q0/s0
Q0/s1
Q0/s2
Q1/s0
Q1/s1
Q1/s2
Q2/s0
Q2/s1
Q2/s2
Target-online IC phase (gate 1.10)
Ratio of in-process ms
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
batch/control median 0.612
Q0/s0
Q0/s1
Q0/s2
Q1/s0
Q1/s1
Q1/s2
Q2/s0
Q2/s1
Q2/s2


N53 K220 reference accounting • 9 verified public Q/seed pairs
Run 20261012T1737Z, shared Apple M4 Pro (L0); dotted = medians, dashed = rho parity
Complete cold method against same-Q strong rho
control/rho median 78.146
batch/rho median 52.586
Ratio of in-process ms
10^2
10^1
10^0
Q0/s0
Q0/s1
Q0/s2
Q1/s0
Q1/s1
Q1/s2
Q2/s0
Q2/s1
Q2/s2
Target-online phase against same-Q strong rho
control/rho median 0.490
batch/rho median 0.378
Ratio of in-process ms
1.2
1.0
0.8
0.6
0.4
0.2
Q0/s0
Q0/s1
Q0/s2
Q1/s0
Q1/s1
Q1/s2
Q2/s0
Q2/s1
Q2/s2